The Bergman Putnam Factor-1/2 Conjecture Is Already Answered

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Status

This is a lightweight literature-status packet. It records that the open-problem signal in Bell–Ferguson–Lundberg's paper *Self-commutators of Toeplitz operators and isoperimetric inequalities* [1] is already answered by a separate cited paper of Olsen and Reguera [2].

Source Question

The source arXiv id is 1211.6937. In the first concluding remark, the authors conjecture the following factor-1/2 improvement of Putnam's inequality for Toeplitz operators with analytic symbol ϕ acting on the Bergman space of a planar domain G:

$$\|[T_\phi^*, T_\phi]\| \leq \frac{\text{Area}(\phi(G))}{2\pi}.$$

The same remark contains an "Added in press" note stating that J.-F. Olsen and M. C. Reguera proved the conjecture in Theorem 1 and Corollary 1 of their paper.

Supporting Answer

The supporting arXiv id is 1305.5193. Olsen and Reguera explicitly state that their sharp Hankel-operator estimate improves the classical Putnam inequality for Bergman-space Toeplitz commutators by a factor of 1/2, answering the Bell–Ferguson–Lundberg conjecture. Their Corollary 1 gives the weighted-domain estimate

$$\|T_\psi^* T_\psi - T_\psi T_\psi^*\|_{A^2_\alpha(\Omega)\to A^2_\alpha(\Omega)} \leq \frac{\|\psi'\|^2_{A^2(\Omega)}}{2+\alpha}$$

for $\alpha \geq -1$, simply connected $\Omega \subset \mathbb{C}$, and analytic ψ. Taking $\alpha = 0$ and applying this in the setting of the source conjecture gives the announced factor-1/2 Bergman Putnam inequality.

Scope

This is not a new solution packet and should not be counted as an original solve. It is an exact literature answer: the source paper itself identifies the supporting paper and says the conjecture was proved there.

References

[1] S. R. Bell, T. Ferguson, and E. Lundberg, *Self-commutators of Toeplitz operators and isoperimetric inequalities*, arXiv:1211.6937, 2012.

[2] J.-F. Olsen and M. C. Reguera, *On a sharp estimate for Hankel operators and Putnam's inequality*, arXiv:1305.5193, 2013.